Pack Opening VFX Animation Specification

Technical reference for the card pack opening animation system. All timings, dimensions, colors, and layer ordering documented below match the current implementation.

Table of Contents

1. Overall Flow & Phase Diagram
2. Dimensions & Aspect Ratios
3. Phase-by-Phase Breakdown
4. Card Flip & Reveal System
5. Rarity Tier Reference
6. Particle & Spark Systems
7. Video Overlay System
8. Holographic Card Effects
9. Z-Index Layer Map
10. Easing Functions

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1. Overall Flow & Phase Diagram

```
ENTER (900ms) → READY (idle) → TEARING (user drag) → RIPPING (5500ms) → EMERGING (3500ms) →
STACK (interactive) → SUMMARY
```

Phase	Duration	Trigger	Description
Enter	900ms	Automatic	Pack flies in from below, scales up
Ready	Indefinite	After enter	Pack floats idle, edge dots pulse, awaits user
Tearing	User-driven	Touch/drag on pack	Tear line progresses with user interaction
Ripping	5500ms	Auto at 95% tear	Pack tears open, light spills from slit
Emerging	3500ms	After ripping	Cards rise out of torn pack, fan out, regroup
Stack	Interactive	After emerging	Cards stacked; user clicks to flip-reveal each one
Summary	500ms entry	After last card	All revealed cards shown in summary grid

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2. Dimensions & Aspect Ratios

Pack

Property	Value	Notes
Aspect ratio	63:100	Portrait orientation
Standard width	190px	Default render size
Compact width	140px	Smaller contexts

Trading Card (Player Cards)

Property	Value	Notes
Aspect ratio	63:88	Standard trading card proportions
Base width	340px	Full-size render
Base height	~474.6px	Derived from aspect ratio
Scaling unit	`cqi`	1 cqi = 1% of container width
Border radius	16px	All corners

Game Card (Items/Abilities)

Property	Value	Notes
Aspect ratio	63:88	Same as trading cards
Base width	240px	Standard size
Compact width	180px	Reduced layout

Scene (Flip Test Page)

Property	Value	Notes
Scene width	340px	
Scene height	476px	
Perspective	1200px	3D camera depth

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3. Phase-by-Phase Breakdown

Phase 1: Enter (900ms)

Property	Start	End	Easing
Scale	0.3	1.0	cubic-bezier(0.34, 1.56, 0.64, 1)
Opacity	0	1	Same
Y offset	+80px	0	Same

Keyframe name: `po-enter`

Phase 2: Ready (Idle)

Float animation — `po-float`, 3s ease-in-out, infinite loop:

- Y translation: 0 → -10px → 0

Edge dot pulse — `po-dot-pulse`, 1.5s infinite:

- Scale: 0.8 → 1.3
- Opacity: 0.3 → 0.9

Phase 3: Tearing (User-Controlled)

- Progress tracked as `--progress` CSS variable (0 to 1)
- Glow along tear line intensifies: `0 0 calc(4px + progress * 10px)`
- Auto-completes when: progress >= 0.95 AND distance traveled >= 0.25

Phase 4: Ripping (5500ms total)

Sub-animation	Duration	Delay	Easing	Description
Pack clip	Instant	0ms	—	`clip-path: inset(9% 0 0 0)` — top 9% splits
Top tear away	1000ms	2000ms	ease-in-out	Y: 0 → -100px, opacity: 1 → 0
Bottom slide down	1400ms	150ms	ease-in	Y: 0 → +400px, opacity: 1 → 0
Shine sweep	2500ms	0ms	cubic(0.12,0,0.39,0)	Horizontal light beam sweeps L→R
Rarity glow build	2500ms	2500ms	ease-out	Cone of light from slit, scaleY: 0.03 → 1

Shine sweep (`po-shine-sweep`): A bright horizontal line sweeps across the pack slit from left to right, followed by a vertical glow spill upward. Represents light escaping from inside.

Rarity glow (`po-glow-build`): Colored cone of light emanates upward from the tear slit. Color matches the rarest card in the pack. Opacity ramps 0 → 0.9.

Phase 5: Emerging (3500ms)

Card emergence — `po-card-emerge` , 3000ms, cubic-bezier(0.22, 0.61, 0.36, 1):

```
  0%   — Cards at Y +15px, opacity 0 (hidden inside pack)
  5%   — Opacity snaps to 1
 25%   — Cards rise to Y -80px (above pack)
 38%   — Hold position above pack
 52%   — Fan out: each card offsets by --fx (X) and rotates by --fr
 66%   — Hold fanned position
 78%   — Fan back together (X → 0, rotation → 0)
 92%   — Begin moving to final center position
 95%   — Z-index jumps from 5 to 30 (cards move in front of pack)
100%   — Cards at Y 0, centered, stacked
```

Accompanying effects during emergence:

- `po-cards-glow-in` (3500ms ease-out): ambient glow behind card stack
- `po-rarest-fan-glow` (3000ms): extra glow pulse on the rarest card during the fan-out moment

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4. Card Flip & Reveal System

Each card in the stack is revealed one at a time via a click → anticipation → flip → reveal sequence.

Anticipation Phase (Pre-Flip)

Starts on click. Duration and intensity scale with rarity:

Rarity	Wait Duration	Shake Animation	Shake Magnitude	Screen Veil Opacity
Common	0ms	None	0	0
Uncommon	0ms	None	0	0
Rare	300ms	None	0	0.15
Epic	600ms	`po-shake-epic` (80ms)	+/-1.5px	0.35
Legendary	1000ms	`po-shake-legendary` (60ms)	+/-2.5px	0.50
Mythic	1400ms	`po-shake-mythic` (40ms)	+/-3px	0.65

Last card bonus: Add +200ms to +800ms to the wait, depending on the tier of the last card. This creates extra tension for the final reveal.

Rising sparks appear during anticipation (see [Particle Systems]).

Card Flip

Rarity	Flip Duration	Last Card Duration	Easing
Common	400ms	600ms	ease-in-out
Uncommon	500ms	700ms	ease-in-out
Rare	500ms	700ms	ease-in-out
Epic	600ms	800ms	ease-in-out
Legendary	700ms	900ms	ease-in-out
Mythic	800ms	1000ms	ease-in-out

Transform: `rotateY(0deg → 180deg)` with backface-visibility hidden on both faces.

Reveal Effects (Fire Simultaneously at Flip Completion)

Screen Flash

Rarity	Animation	Duration	Peak Opacity	Color
Rare	`po-flip-flash`	400ms	0.70	White
Epic	`po-flip-flash-tinted`	500ms	0.70	#a855f7 (purple)
Legendary	`po-flip-flash-tinted`	600ms	0.70	#ff8c00 (orange)
Mythic	`po-flip-flash-big`	600ms	0.95→0.6→0	#ef4444 (red)

Easing: ease-out. Flash covers the full viewport.

Radial Glow (Behind Card)

Animation: `po-radial-glow-in`, 800ms ease-out

- Scale: 0.5 → 1.0
- Opacity: 0 → 1
- Inset: -120px (standard), -240px (Legendary+)
- Color: rarity color, radial gradient fading to transparent

Mythic-only rotating rays:

- Conic gradient mask, infinite rotation at 8s/revolution (`po-rays-spin`)
- Entry animation: `po-rays-enter`, 800ms ease-out

Ring Burst (Epic+ Only)

Variant	Animation	Duration	Scale Range	Peak Opacity
Standard	`po-ring-expand`	700ms	0.3 → 3.0	0.7 → 0
Big (Leg/Myth)	`po-ring-expand-big`	900ms	0.2 → 4.0	0.9→0.5→0

Easing: ease-out. Ring is a circular border that expands outward from the card center.

Light Sweep (Across Card Face)

Animation: `po-card-sweep`

- Standard: 500ms ease-out, 100ms delay
- Higher rarities: 700ms
- Gradient sweeps from X -120% to X +120%
- Brightness: 35-50% at center, 15% at edges

Card Lift (3D Depth Pop)

Card translates forward on Z-axis after reveal:

Rarity	TranslateZ	Transition
Rare	8px	400ms ease-out
Epic	12px	400ms ease-out
Legendary	18px	400ms ease-out
Mythic	24px	400ms ease-out

Holographic Shimmer (Post-Reveal Sweep)

Animation: `po-holo-sweep`

- Rare: 2000ms, 12% peak opacity
- Epic: 2500ms, medium opacity
- Legendary/Mythic: 3000ms + pulse (`po-holo-pulse`, 1500ms alternate, starts at 2s)
- All: 150ms delay, ease-in-out

Rarity Banner (Text Slam)

Appears after flip completes to announce the rarity tier.

Rarity	Animation	Duration	Scale Bounce
Rare+	`po-banner-slam`	500ms	1.8→0.95→1.0
Mythic	`po-banner-slam-mythic`	700ms	2.2→0.9→1.08→0.98→1.0

Easing: cubic-bezier(0.16, 1, 0.3, 1) — elastic/overshoot feel. Blur: 8px → 0 during slam.

Display duration before fade:

- Rare+: 1000ms
- Legendary: 1500ms
- Mythic: 2000ms

Typography:

- Font size: 48px (56px for Mythic)
- Letter-spacing: 12px (16px for Mythic)
- Text-shadow: triple glow layers in rarity color

Post-Reveal Continuous Glow

Revealed cards pulse infinitely while visible:

Rarity	Animation	Period	Shadow Range
Rare	`po-revealed-pulse-epic`	2.0s	0 0 20px 5px ↔ 0 0 35px 8px
Epic	`po-revealed-pulse-epic`	2.0s	0 0 35px 8px ↔ 0 0 50px 14px
Legendary	`po-revealed-pulse-legendary`	1.8s	Double-layer shadow, larger
Mythic	`po-revealed-pulse-mythic`	1.5s	Multi-layer massive glow

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5. Rarity Tier Reference

Core Colors

Tier	Rarity	Primary HEX	RGB	Usage
0	Mythic	#ef4444	239, 68, 68	Glow, flash, particles
1	Legendary	#ff8c00	255, 140, 0	Glow, flash, particles
2	Epic	#a855f7	168, 85, 247	Glow, flash, particles
3	Rare	#3b82f6	59, 130, 246	Glow, particles
4	Uncommon	#22c55e	34, 197, 94	Border only
5	Common	#6b7280	107, 114, 128	Border only

Extended Card Border Palette

Each rarity has 4 color variants used for card frame rendering:

Rarity	Primary	Light	Dark	Edge
Common	#c9a84c	#e0c56a	#8a7030	#f0d878
Uncommon	#22c55e	#4ade80	#16a34a	#6ee7a0
Rare	#3b82f6	#60a5fa	#2563eb	#93c5fd
Epic	#a855f7	#c084fc	#7c3aed	#d8b4fe
Legendary	#ff8c00	#ffaa33	#cc7000	#ffc966
Mythic	#ef4444	#f87171	#dc2626	#fca5a5
Unique	#e8e8ff	#ffffff	#c0c0e0	#ffffff

Pack Tier Color Schemes

Pack Tier	Accent	Light Accent	Dark Accent
Standard	#6a7a8a	#a0b0c0	#3a4a5a
Premium	#2563eb	#7db4ff	#1a3a8a
Elite	#7c3aed	#b888ff	#4a1a8a
Legendary	#d97706	#fcd34d	#78350f
Mixed	#6366f1	#a5b4fc	#3730a3
OSL Mixed	#d2b138	#f0d878	#8a7420
BSL Mixed	#2795cf	#6bc0f0	#155a7a
Gift	#dc2626	#f87171	#7f1d1d

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6. Particle & Spark Systems

Anticipation Rising Sparks

Spawn during the anticipation wait before each card flip. Rise upward from the card.

Rarity	Spawn Interval	Sparks/Spawn	Rise Duration	Y Displacement
Rare	180ms	1	0.6–1.2s	-120px to -300px
Epic	120ms	2	0.6–1.2s	-120px to -300px
Legendary	80ms	3	0.6–1.2s	-120px to -300px
Mythic	Varies	Varies	0.6–1.2s	-120px to -300px

Spark properties:

- Size: 2–5px
- X drift: +/-100px (random)

- Scale: 1.2 → 0.2 (shrink as they rise)
- Opacity: 0 → 1 → 0.5 → 0
- Color: rarity color blended with white
- Glow: box-shadow in rarity color + outer halo
- Easing: cubic ease-out trajectory

Keyframe: `po-anticipation-rise`

Reveal Particle Burst

Fires outward from center at the moment of reveal:

Rarity	Particle Count	Spread Radius	Duration Range
Rare	8	130px	500–1100ms
Epic	15	170px	500–1100ms
Legendary	25	220px	500–1100ms
Mythic	40	280px	500–1100ms

Particle properties:

- Easing: cubic-bezier(0.16, 1, 0.3, 1) — elastic out
- Color: 70% rarity color, 30% white
- Direction: radial outward from center, evenly distributed with random jitter
- Size: 3–8px
- Shape: circle with glow shadow

Test Page Particles (Standalone)

Used in the video test harness:

- Count: 20 per burst
- Size: 3–8px
- Color: #f0d060 (gold)
- Glow: 0 0 8px #f0d060
- Spread: 150–350px radius
- Duration: 600–1200ms
- Easing: cubic-bezier(0, 0.6, 0.3, 1)

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7. Video Overlay System

The reveal system supports composited `.webm` video overlays rendered on top of the card for premium glow effects. Currently in testing with three variants.

Container Setup

```
Position:   absolute, inset -40% (extends 40% beyond card edges in all directions)
Blend mode: screen (additive — black = transparent, bright = visible)
Z-index:    10 (above card face, below UI)
Pointer:    none (click-through)
```

The video is 180% the card dimensions (card + 40% bleed on each side).

Video Requirements

Property	Value / Requirement
Format	WebM (VP9 or AV1 codec)
Alpha channel	Not required (screen blend makes black transparent)
Aspect ratio	Square or wider than 63:88 (will be `object-fit: cover`)
Looping	Yes — video loops continuously
Audio	Muted by default (audio track optional)
Background	Dark/black areas become invisible via screen blend

Playback Lifecycle

1. Video is preloaded but hidden (opacity: 0)
2. On reveal: `currentTime = 0`, `.play()`, opacity snaps to 1
3. After 4000ms: opacity transitions to 0.5 over 2000ms (gradual fade)
4. On reset: pause, opacity → 0 (300ms transition)

Test Videos (Currently Staged)

File	Description
`OSLPackTest1.webm`	VP9 encoded test
`OSLPacktest2_EM_T.webm`	AV1 encoded test (transparency?)
`OSLpacktest3_IA.webm`	VP9 with audio track

Design Guidelines for New Videos

- **Bright areas glow**, dark areas vanish (screen blend mode)
- Content should be centered — the card sits in the middle of the video frame
- Edge effects (rays, pulses, particles) work best since they extend beyond the card
- Aim for 3–6 second loops with seamless loop points
- Keep the first frame dark/black so there's no flash on load

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8. Holographic Card Effects

The holo system is a multi-layer shader-like effect driven by mouse/pointer position. These are the post-reveal interactive effects on revealed cards.

Pointer-Tracking CSS Variables

```
--pointer-x            Radial center X (percentage)
--pointer-y            Radial center Y (percentage)
--pointer-from-left    0 (left edge) → 1 (right edge)
--pointer-from-top     0 (top edge) → 1 (bottom edge)
```

```
--pointer-from-center  0 (center) → 1 (corner)
--background-x         Background offset X
--background-y         Background offset Y
```

3D Tilt (Mouse Tracking)

- RotateX: `(0.5 - pointerY) * 15deg`
- RotateY: `(pointerX - 0.5) * 20deg`
- Perspective: 600px
- Transition: 0.15s ease-out (pack), 0.4s ease (card box-shadow)
- Transform-style: preserve-3d

Layer Stack (Front of Card)

Layer	Z	Blend Mode	Description
Card	0	normal	Base card art and frame
Shine	2	color-dodge / soft-light	Primary holo pattern
Shine2	2	(Unique rarity only)	Secondary overlay
Glare	3	overlay / hard-light	Specular highlight at pointer

Holo Clip Paths

Effects can be masked to only the image area or everything except the image:

Holo mode (shine on image only):

- Player card: `inset(8.6% 5.3% 47% 5.3%)`
- God card: `inset(15% 6% 53% 6%)`

Reverse mode (shine on frame, not image):

- Polygon with evenodd fill-rule cutting out the image rect

Holo Variants by Rarity

Rarity	Effect Name	Key Visual
Common	None	No shine layer (hidden)
Uncommon	Holo Standard	Rainbow striped gradient, color-dodge
Rare	Glitter	Subtle sparkle texture, radial glow
Epic	Secret Gold	Double glitter + conic rainbow + gold accent
Legendary	Gold Sparkle	Gold-focused spectrum, geometric pattern
Mythic	Radiant Crosshatch	45°/-45° crosshatch + rainbow band overlay
Unique	Radiant + Rainbow	Double-stacked: crosshatch + rainbow spectrum

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9. Z-Index Layer Map

Pack Opening Scene

```
z:    1  — Background grid / scene backdrop
z:    5  — Emerging cards (during pack-clearing phase)
z:   10  — Pack area + tear zone
z:   14  — Rarity glow (cone from slit)
z:   20  — Pack sparkles + torn top piece
z:   22  — Shine sweep (light beam)
z:   24  — Screen veil (anticipation dimming)
z:   25  — Emerging cards container
z:   30  — Cards in stack phase + tear zone overlay
z:   35  — Reveal particles (burst)
z:   36  — Anticipation sparks (rising)
z:   60  — Screen flash (full viewport)
z:   61  — Rarity banner text
z:  999  — Replay button
z: 1000  — Zoom overlay (full-screen card view)
```

Pack Art (Store Display)

```
z:  0  — Foil base layer
z:  1  — God art (full bleed)
z:  2  — Foil overlay + specular bands
z:  3  — Grain texture
z:  5  — Bottom metallic band
z:  6  — Logo, badges, title, card count
z:  9  — Glare (mouse-tracked)
z: 10  — Shimmer sweep
z: 11  — Edge bevel
z: 15  — Crimp edges (top/bottom)
```

Video Overlay (Test Page)

```
z:  -1  — Card glow (behind card)
z:   5  — Holo sweep (on card face)
z:  10  — Video glow overlay
z:  15  — Sparkle particles
z:  20  — Ring burst
z: 100  — Screen flash
z: 200  — UI controls
```

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10. Easing Functions

Name / Use	Value	Character
Pack enter	`cubic-bezier(0.34, 1.56, 0.64, 1)`	Overshoot/bounce
Shine sweep	`cubic-bezier(0.12, 0, 0.39, 0)`	Slow start, fast end
Card emergence	`cubic-bezier(0.22, 0.61, 0.36, 1)`	Smooth deceleration

Banner slam	`cubic-bezier(0.16, 1, 0.3, 1)`	Elastic/springy
Particle burst	`cubic-bezier(0.16, 1, 0.3, 1)`	Elastic out
Test page particles	`cubic-bezier(0, 0.6, 0.3, 1)`	Fast start, gentle end
Card flip	`ease-in-out`	Standard
Ring expand	`cubic-bezier(0, 0.5, 0.3, 1)`	Quick start, slow end
3D tilt response	`ease-out` (0.15s)	Responsive follow
Glow transitions	`ease-out`	Natural fade

Appendix: Role Color Palettes (Card Backgrounds)

Each player role has a dark body + accent color scheme:

Role	Accent Color	HEX
ADC	Bronze/amber	#c4884a
Solo	Steel blue	#4a8ec4
Jungle	Forest green	#4ab868
Mid	Mystic purple	#9a5cc4
Support	Dark teal	#4aaab8

Source files: `src/pages/vault/components/PackOpening.jsx`, `PackOpening.css`, `PackArt.jsx`, `PackArt.css`, `GameCard.css`, `src/components/TradingCard.jsx`, `TradingCard.css`, `TradingCardHolo.jsx`, `TradingCardHolo.css`, `public/test-video.html`